Technical Specification: Petrol — LLM-Assisted Procurement Report Summarization

Document type: Internal technical specification **Project:** Petrol ProcureAssist (internal codename) **Authors:** Maša Vidmar, Rok Lampret **Date:** 2025-01-14 **Version:** 0.3 — draft, not yet reviewed by client **Status:** In progress — do not distribute

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1. Background

Petrol's procurement team generates and receives approximately 120–150 supplier performance reports per month across their fuel, energy services, and EV charging divisions. These reports arrive in PDF format (structured and unstructured), Excel summaries, and occasionally as plaintext email attachments. Each report is 8–35 pages.

A procurement analyst currently spends 2–3 hours per report extracting key metrics, flagging anomalies, and populating a master Excel tracker. Petrol has asked Kvant to build a pipeline that automates the initial extraction and summary step, reducing analyst time to a review-and-correct workflow.

Contract value: €28,500 **Project start:** January 20, 2025 **Estimated delivery:** March 31, 2025

This spec covers the technical design for the extraction and summarization pipeline. It does not cover the UI layer (separate spec, TBD by Rok) or the integration with Petrol's SAP system (out of scope for phase 1).

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2. Input Document Taxonomy

Based on the 40-document sample Petrol shared in December 2024:

Document type	Format	Frequency/month	Avg. pages	Structured?
Supplier KPI reports	PDF (text-layer)	45–55	12	Partially (tables + narrative)
Fuel delivery summaries	Excel (.xlsx)	30–40	N/A	Yes
Compliance certificates	PDF (often scanned)	15–20	3–5	No

Document type	Format	Frequency/month	Avg. pages	Structured?
Incident/deviation reports	PDF or email body	10–15	2–8	No
Quarterly business reviews	PDF (text-layer)	8–12	25–35	Partially

Key finding from document audit: Approximately 22% of incoming PDFs are scanned images (no text layer). These require OCR preprocessing. Petrol was not aware this was the case — their suppliers apparently scan signed copies without running OCR. This has cost implications (see §6).

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3. System Architecture

3.1 Pipeline Overview

```
[Input documents]
    → [Ingestion & classification]
    → [Preprocessing (OCR if needed, table extraction)]
    → [LLM extraction pass]
    → [Structured output (JSON)]
    → [Human review UI]
    → [Export to Petrol tracker (Excel/CSV)]
```

3.2 Component Specifications

Ingestion and Classification

- Trigger: file drop to monitored S3-compatible bucket (Hetzner Object Storage)
- Classification model: fine-tuned or few-shot prompted GPT-4o-mini to categorize by document type (5 categories above)
- Confidence threshold: if classification confidence < 0.85, route to human review queue
- Expected classification accuracy: >92% on held-out sample (to be validated)

Preprocessing Text-layer PDFs: - PyMuPDF for text extraction - Camelot or pdfplumber for table extraction (to be evaluated — Camelot better on simple grids, pdfplumber more robust on complex layouts) - Chunk preservation: tables extracted as structured JSON, not flattened to text

Scanned PDFs: - AWS Textract for OCR (Petrol already has AWS account; cheaper than standing up Tesseract pipeline) - Cost: approximately

€0.0015/page for Textract Analyze Document - At 15–20 scanned docs/month, avg 4 pages = 60–80 pages/month €0.09–0.12/month (negligible)

Excel files: - openpyxl for parsing - Heuristic column detection for KPI tables (supplier name, delivery volume, on-time %, defect rate, penalty flags)

LLM Extraction Pass This is the core of the system. For each document (after preprocessing), we send structured prompts to GPT-4o to extract a normalized JSON object.

Target extraction schema (KPI report example):

```
{
  "document_type": "supplier_kpi_report",
  "supplier_name": "string",
  "report_period": "YYYY-MM",
  "kpis": [
    {
      "metric_name": "string",
      "value": "number or string",
      "unit": "string",
      "target": "number or null",
      "status": "on_target | below_target | above_target | unknown"
    }
  ],
  "anomalies_flagged": ["string"],
  "summary_narrative": "string (2-4 sentences)",
  "confidence_score": "float 0-1",
  "requires_human_review": "boolean"
}
```

Prompt strategy: - System prompt: role definition + schema + extraction rules - User prompt: preprocessed document text/tables - For long documents (>6,000 tokens of content): sliding window with overlap, merge extraction results - Temperature: 0 (extraction task, want determinism) - Model: GPT-4o (not mini — Rok tested mini on the sample set, structured output reliability was poor on complex tables)

Validation Layer After extraction, a rule-based validation pass checks: - Required fields are present - Numerical values are within plausible ranges (configurable per document type) - Report period is a valid date and not in the future - Supplier name matches known supplier list (fuzzy match against Petrol's supplier registry — CSV provided by Petrol)

Validation failures trigger human review flag.

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4. Human Review Interface

Basic Streamlit app (internal access only): - Shows extracted JSON alongside original document (side-by-side PDF viewer) - Editable fields for analyst to correct extraction errors - Approve / Send back / Escalate actions - Correction data logged for future fine-tuning consideration

This is not a priority feature — Petrol wants the pipeline working first. UI can be rough.

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5. Output and Integration

Phase 1 output: CSV/Excel export of extracted structured data, formatted to match Petrol's existing tracker columns (mapping provided by Petrol procurement team, December 18 email).

Phase 2 (out of scope): Direct SAP integration via API. Petrol IT has indicated this would require 6–8 months of their internal process and is not feasible for the current project timeline. We should document this clearly in the contract so it doesn't become a phase 1 expectation.

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6. Cost and Infrastructure

Infrastructure (monthly, steady state)

Item	Cost (EUR/month)
Hetzner CX21 (pipeline server)	€8.90
Hetzner Object Storage (input bucket)	€3–5
AWS Textract (OCR, scanned docs)	<€1
OpenAI API (GPT-4o, ~150 docs/month)	€80–130 (est.)
Total	**~€95–145/month**

Note: OpenAI cost estimate assumes average 3,500 tokens input + 800 tokens output per document, at current GPT-4o pricing (€2.50/1M input, €10/1M output as of January 2025). This will vary.

One-time development costs (internal tracking)

Phase	Engineer	Days	Internal cost
Ingestion + classification	Rok	5	€3,500
Preprocessing (PDF/OCR/Excel)	Rok + Nika	7	€5,600
LLM extraction + schema design	Maša	6	€5,400

Phase	Engineer	Days	Internal cost
Validation layer	Rok	3	€2,100
Human review UI	Luka	4	€2,000
Testing + iteration	Maša + Rok	5	€4,500
DevOps + deployment	Matic	3	€2,400
PM + client coordination	Eva	4	€2,800
Total internal		**37 days**	**€28,300**

Margin on this project is thin (€28,500 contract vs. ~€28,300 costs). The OCR discovery is the main culprit — we didn't price it in. Maša flagged this in the November estimate; Tine agreed to absorb it to keep the relationship. Worth watching the actual hours.

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7. Open Issues

- ☐ Camelot vs. pdfplumber decision — Rok to test on 5 sample documents by January 20
- ☐ Petrol supplier registry CSV — requested December 18, not yet received
- ☐ Confirm AWS Textract permissions with Petrol IT (they have the account but we need API access)
- ☐ Long document handling (QBRs, 25–35 pages) — sliding window approach needs testing, may need different strategy
- ☐ What happens when a supplier submits a report in a format we haven't seen? Need fallback/routing logic

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8. Notes and Concerns

The scanned PDF percentage (22%) is higher than expected and is the biggest technical risk. If Textract performs poorly on low-quality scans (which is possible — some of the samples Petrol shared look like fourth-generation photocopies), we'll need a human fallback that could eat into time savings.

Rok is doing a quality check on the OCR output this week. If accuracy is below 85% on the test set, we need to have a conversation with Petrol about document quality standards before we go live.

Also: Petrol has mentioned several times that they eventually want this to "integrate with everything." SAP, their internal BI dashboards, Teams notifications. We need to hold the line on phase 1 scope. Eva is drafting a scope confirmation letter — please review before sending.